AGROBIOMETRIX: USAID DEVELOPMENT INNOVATION VENTURES (DIV)

Stage 1 (Piloting) Complete Grant Application Dossier, Master Proposal Responses & Scientific Collaboration Guide

Applicant Org: Naysa-84 Labs, Nigeria	**Principal Investigator:** Dr. Young I. Irivboje	**Funding Stream:** USAID DIV Stage 1	**Requested Amount:** $150,000 USD
Project Duration: 2 Years (24 Months)	**Pathway to Scale:** Hybrid Model	**Evaluation Type:** Quasi-Experimental (DiD)	**Target Geography:** Nigeria (Kano, Oyo/Ogun)

1. Project Title & Executive Summary

Prompt: Project Title [up to 250 characters]

AgroBiometrix: Bridging the Last-Mile Agricultural Extension Gap Through AI-Powered Biometrics, Community FM Radio Syndication, and Vernacular 2G GSM Voice Advisories for Smallholder Farmers in Nigeria (209 characters)

Prompt: Application summary [up to 650 characters]

AgroBiometrix is an agricultural biometrics platform that translates scientific crop, soil, and climate data into localized vernacular audio (Hausa, Yoruba). It serves resource-poor Nigerian smallholder farmers with basic 2G phones and zero internet, alongside public researchers. It aims to increase crop yields by 15–25%, prevent chemical fertilizer wastage, and raise household farm income. With DIV funding, we will execute a 2-year quasi-experimental pilot across 3,000 farmers, syndicating bulletins via community FM radio and automated 2G GSM robocalls to generate rigorous causal evidence on advisory adoption and productivity gains. *(641 characters)*

2. Problem Statement

Prompt: What specific development challenge are you addressing, and who does it affect? [up to 975 characters]

Sub-Saharan Africa faces a severe agricultural yield crisis driven by a systemic last-mile extension failure. While research institutes develop high-yielding, climate-smart varieties and optimal soil formulas, over 80% of smallholders never receive actionable guidance. Public extension is critically understaffed (a 1:8,000 agent-to-farmer ratio in Nigeria), while commercial digital agtech requires smartphones, broadband data, and English literacy—tools rural smallholders lack.

Consequently, poor farmers suffer 30–50% preventable crop yield deficits, misallocate scarce income on unbalanced fertilizers, and remain vulnerable to climate shocks. Women farmers, who face greater literacy barriers, are disproportionately affected.

Affected Populations:
(i) Target Country (Nigeria): ~38 million smallholders (supporting over 100 million rural dependents).
(ii) Globally: ~500 million smallholder farming families (>2 billion people). *(938 characters)*

3. Core Innovation & Comparative Differentiation

Prompt: What is the core innovation, and how does it compare to existing alternatives? [up to 2,300 characters]

(i) What It Is and How It Works:
AgroBiometrix is an agricultural biometrics operating system coupled with an automated last-mile vernacular voice dissemination engine. It bridges the gap between scientific research and resource-constrained smallholders. The platform features 8 modular suites that process complex trial data, including mixed-model ANOVA, GxE AMMI/GGE stability biplots, stoichiometric soil nutrient balancing (lime and NPK requirements), and bioclimatic livestock heat stress (THI). A multi-agent AI engine translates these statistical outputs into spoken indigenous audio (Hausa, Yoruba, Swahili, French).

Typical Use Case Walkthrough:
• *At the Research Station:* An agronomist or extension supervisor at a national institute (such as IAR or ARCN) uploads regional crop trial and soil sample data. AgroBiometrix computes site-specific planting dates, fertilizer recommendations, and pest thresholds, generating vernacular audio bulletins.
• *At the Farm Level:* Amina, a non-literate maize smallholder in Kano with a basic $10 2G feature phone, receives advisories through two channels: (1) Community FM Radio, where her local station broadcasts our automated RSS podcast feed during morning farm hours; and (2) Direct 2G GSM Robocalls, where the system dials her phone during urgent pest outbreaks or planting windows, delivering a 1-minute voice advisory in native Hausa. No smartphone, internet, or literacy is required.

(ii) Comparison to Existing Alternatives:
1. *In-Person Extension:* Traditional extension is severely understaffed (1:8,000 agent-to-farmer ratio in Nigeria) and expensive ($50-$100 per visit). AgroBiometrix achieves mass reach at fractions of a cent per farmer.
2. *Smartphone AgTech Apps:* Most digital apps require 4G smartphones, mobile data, and English literacy, excluding over 80% of African smallholders. AgroBiometrix operates over FM radio and 2G cellular voice.
3. *Text SMS Advisories:* SMS fails for non-literate farmers. AgroBiometrix uses indigenous tonal voice synthesis. *(2,013 characters)*

4. Theory of Change & Causal Impact Chain

Prompt: What key development outcomes do you expect your innovation to improve, and how? [up to 2,300 characters]

(i) Key Measurable Outcomes:
1. *Crop Productivity:* A 15% to 25% increase in staple crop yields (maize, sorghum, cassava) per hectare among participating smallholders.
2. *Input Cost Efficiency:* A 20% to 30% reduction in misallocated chemical fertilizer spending through precision stoichiometric lime and NPK balancing, preventing nutrient lock-out.

3. *Household Income & Living Standards:* An estimated $80 to $150 annual increase in net agricultural income per household, representing a 15% to 20% increase in financial well-being for families living under $2/day.
4. *Climate Resilience:* Reduced livestock heat stress mortality and crop failure rates during acute dry spells.
These outcomes directly overcome the productivity trap caused by the collapse of public extension.

(ii) Theory of Change & Causal Chain:
1. *Activity (Data Ingestion & Synthesis):* Research institutes and universities (such as ARCN and IAR) upload regional trial data and soil samples into AgroBiometrix. The platform runs local biometric models, calculating optimal planting dates, stoichiometric fertilizer formulas, and pest thresholds.
2. *Knowledge Access Mechanism:* Rather than producing complex English manuals, AgroBiometrix synthesizes recommendations into native spoken audio (Hausa, Yoruba). It disseminates advisories via community FM radio syndication and automated 2G GSM phone robocalls directly to basic feature phones.
3. *Behavioral Change:* Smallholder farmers who previously lacked access to extension now receive clear, timely instructions in their mother tongue. Farmers shift from guesswork to precision practices: calibrating sowing to real moisture windows, applying balanced lime/NPK, and treating pest outbreaks early.
4. *Final Impact:* Increased yields and avoided input waste raise net farm revenues. This causal mechanism is validated by existing development economics literature (e.g., Cole & Fernando, 2021; Duflo et al., 2011), which demonstrates that timely voice-based agricultural advisories increase adoption of optimal inputs by 10 to 18 percentage points, directly driving household income growth. *(2,097 characters)*

5. Proposed Use of DIV Funding & Budget Breakdown

Prompt: What specific activities will the DIV Fund grant support? [up to 2,000 characters]

Catalytic Role of DIV Funding:
While AgroBiometrix's software is fully operational, without DIV funding it would remain confined to research stations. DIV provides the catalytic capital to deploy the technology in the field, financing real-world pilot testing across 3,000 smallholders and generating rigorous causal evidence of its impact.

Major Supported Activities ($150,000 Total Requested):
• **1. Farmer Enrollment & Cluster Onboarding (~30% / $45,000):** Register 3,000 smallholders across 30 rural clusters in two agro-ecological zones (Kano and Oyo/Ogun). Builds the operational testing base and establishes treatment and control cohorts for field testing.
• **2. Impact Evaluation & Evidence Generation (~27% / $40,000):** Execute baseline and endline household surveys across treatment and control groups, led by our agricultural economics team. Generates rigorous Difference-in-Differences evidence on advisory adoption, yield gains, and cost-effectiveness for Stage 2 scaling.
• **3. Dissemination Float & Radio Syndication (~23% / $35,000):** Fund Africa's Talking 2G airtime for ~50,000 outbound phone robocalls and sponsor weekly syndication across 3 community FM stations. Tests and optimizes last-mile delivery to smallholders lacking smartphones or internet.
• **4. Dialect AI & NARS Agronomic Calibration (~10% / $15,000):** Fine-tune neural voice models to local dialects (Kano Hausa, Oyo Yoruba) and calibrate soil/crop algorithms with ARCN/IAR field trials. Maximizes advisory comprehension and scientific precision.
• **5. Project Coordination, Travel & Compliance (~10% / $15,000):** Oversee field logistics, train enumerators, and manage financial reporting. Ensures milestone execution on time and within budget. *(1,803 characters)*

Milestone Activity Area	Key Functional Deliverable	Share (%)	Amount ($USD)
1. Farmer Enrollment & Clusters	3,000 smallholders registered across 30 rural clusters	30%	**$45,000**
2. M&E; & Impact Evidence	Baseline & endline surveys, DiD causal econometric modeling	27%	**$40,000**
3. 2G Robocalls & FM Radio	~50,000 outbound cellular robocalls + 3 radio station syndications	23%	**$35,000**
4. Dialect AI & NARS Calibration	Hausa/Yoruba speech tuning + IAR/ARCN field trial grounding	10%	**$15,000**
5. Project Travel & Compliance	Field supervisor stipends, travel, and financial audit reporting	10%	**$15,000**
TOTAL PROPOSED BUDGET	**Stage 1 Pilot Execution over 2 Years (24 Months)**	**100%**	**$150,000**

6. Evidence of Impact to Date

Prompt: What existing evidence demonstrates that your innovation improves key development outcomes? [up to 2,000 characters]
As a Stage 1 applicant, causal impact evidence for AgroBiometrix is being generated through this proposed pilot. Our expected impact is grounded in rigorous external causal evaluations of voice-based extension and our internal operational feasibility testing:

1. External Causal Evidence on Voice-Based Extension:
• *Cole & Fernando (2021) [Randomized Controlled Trial, N=1,200]:* Evaluated mobile voice advisories for smallholders. The RCT demonstrated that automated voice advisories increased adoption of recommended agronomic practices by 10.5 to 17.5 percentage points, raised crop yields by 15% to 22%, and yielded a 53% economic return on investment.
• *Fabregas, Kremer, & Schilbach (2019) [Meta-Analysis of RCTs across Sub-Saharan Africa, Science]:* Found digital agricultural advisories increased adoption of recommended inputs (including lime and fertilizer) by 22%, with exceptionally high cost-effectiveness.
• *Van Campenhout et al. (2021) [RCT, IFPRI]:* Found localized vernacular audio advisories increased input adoption by 14 percentage points and staple yields by 16%, with the largest gains among non-literate female smallholders.

2. AgroBiometrix Feasibility & Pilot Data to Date:
• *Technical Feasibility:* All 8 intelligence suites are fully functional and bench-tested locally using historical empirical datasets from Nigerian research trials (maize, cassava, and cattle biometrics from FUNAAB and NARS).
• *Linguistic Intelligibility:* Pre-pilot listening evaluations with native Hausa and Yoruba speakers demonstrated over 90% comprehension of synthesized agronomic terms and pest thresholds.

• *Infrastructure Testing:* Confirmed end-to-end functionality of automated RSS 2.0 podcast feeds for community FM radio and successful API voice dispatch via Africa's Talking 2G GSM cellular gateway. *(1,809 characters)*

7. Delivery Costs & Potential Cost-Effectiveness

Prompt: Current unit cost and projected unit cost at scale [up to 2,000 characters]

(1) Current Unit Cost (Pilot Stage: 3,000 Smallholders over 2 Years):

We distinguish between the one-time research pilot cost and the operational delivery cost:

• *Fully Loaded Pilot Cost:* ~$25.00 per farmer per year ($50 total over 2 years). This is driven by one-time research investments: field enumerator stipends, pre-planting baseline and post-harvest endline household surveys, and cluster onboarding ($85,000 across 3,000 farmers).

• *Operational Tech Delivery Cost:* ~$1.20 per farmer per year.

Calculation & Cost Drivers:

- Fixed Infrastructure: Cloud server ($240/yr), Gemini AI reasoning ($120/yr), and African voice AI synthesis ($180/yr) totaling $600/yr ($0.20/farmer/yr at 3,000 scale).
- Variable Delivery: Africa's Talking 2G cellular voice airtime at ~$0.0075 per 1-minute robocall (12 seasonal alerts/year = $0.09/farmer/yr) plus community radio syndication (~$0.90/farmer/yr). Costs are based on verified vendor tariff data and field staffing projections.

(2) Projected Unit Cost at Scale (100,000 to 1,000,000 Smallholders):

At scale, unit costs collapse because software and radio broadcasts exhibit near-zero marginal cost per additional user:

• *At 100,000 Enrolled Smallholders (Direct 2G GSM Robocalls):* Unit cost falls to ~$0.15 to $0.25 per farmer per year. Bulk telecom carrier agreements reduce 2G voice airtime by 40% (to ~$0.004/call), while fixed cloud and AI costs ($600/yr) dilute to $0.006 per farmer.

• *At 1,000,000+ Smallholders (Community FM Radio Syndication):* Unit cost falls below $0.02 (2 cents) per listener per year. Automated RSS podcast feeds transmit over open FM airwaves to regional farming populations without per-user cellular charges.

Primary cost driver: Cellular airtime, which declines significantly with wholesale volume agreements. *(1,779 characters)*

Prompt: Potential cost-effectiveness [up to 1,300 characters]

At scale, AgroBiometrix delivers unprecedented cost-effectiveness, generating an estimated $160 to $400 in net smallholder income for every $1 spent.

Cost-per-Outcome Estimate at Scale:

• Unit Delivery Cost: ~$0.20 per farmer/year for direct 2G GSM robocalls, and <$0.02 per farmer/year via community FM radio syndication.
• Expected Net Outcome: A 15% to 20% crop yield increase and avoided input waste, producing ~$80 in net household income gains annually.
• Impact per Dollar: Delivering an $80 income gain at an operating cost of $0.20 to $0.50 yields an extraordinary benefit-to-cost ratio exceeding 160:1.

Why It Outperforms Alternatives:

1. *In-Person Extension:* Public extension costs $50 to $100 per farmer visit due to vehicle, fuel, and per diem logistics, achieving a return ratio of barely 1.3:1. AgroBiometrix eliminates field travel costs completely.
2. *Blanket Fertilizer Programs:* Governments spend billions on subsidies, yet farmers misapply inputs. AgroBiometrix's stoichiometric balancing saves $30 to $50 annually per farmer by preventing nutrient waste before planting begins.
3. *Near-Zero Marginal Cost:* Radio airwaves and automated cellular voice APIs scale across millions of farmers without requiring additional field staff. *(1,215 characters)*

8. Scale & Long-Term Financial Sustainability

Prompt: Current scale, anticipated scale, and sustainability [up to 2,000 characters]

(1) Current Scale:

As a Stage 1 post-prototype innovation, AgroBiometrix has to date reached approximately 150 agricultural researchers, university faculty, postgraduate students, and extension officers across partnering institutions (FUNAAB, Federal University of Technology Ilaro, and OOU). Users accessed the digital platform via our web interface during alpha bench-testing and biometric model calibration with empirical trial datasets (maize, cassava, and cattle). Access was provided 100% free, supported by academic research efforts.

(2) Anticipated Scale:

• *Grant Period (2 Years):* We will directly reach 3,000 smallholder farmers across 30 rural clusters receiving targeted 2G GSM robocall advisories and participating in baseline/endline surveys. Concurrently, an estimated 75,000 to 100,000 smallholders will be reached via weekly syndicated vernacular bulletins across 3 community FM radio stations in pilot corridors.

• *At Scale (10-Year Horizon):* We project reaching over 1,500,000 smallholders across Sub-Saharan Africa. Projections are grounded in rural radio penetration (over 75% in agrarian belts) and mobile cellular teledensity exceeding 85% in Nigeria.

Long-Term Sustainability (Hybrid Model):

Smallholders access advisories completely free. Long-term costs are sustained through:

1. *Commercial B2B Revenue:* Seed breeders, input distributors, and agribusinesses pay annual software licensing fees for the biometric and soil modules (Suites 2, 4, 5) and sponsor verified vernacular radio advisory bulletins.
2. *Public Sector Institutionalization:* Adoption by the Agricultural Research Council of Nigeria (ARCN/NARS) and State Agricultural Development Programs (ADPs) as their national digital extension platform, aligned with Nigeria's National Agricultural Technology and Innovation Policy (NATIP). *(1,822 characters)*

9. Evidence, Learning & Impact Evaluation Plan

Prompt: Evidence and learning questions [up to 1,300 characters]

We will answer three key learning questions during the 2-year pilot:
1. Causal Advisory Adoption: Does delivering localized biometrics via native vernacular audio causally increase smallholder adoption of precision agronomy (planting timing, stoichiometric lime/NPK, early pest control) compared to traditional extension?
• *Why Critical:* Tests our core Theory of Change before scaling.
• *Method:* Cluster-level Difference-in-Differences (DiD) surveys across 3,000 farmers (treatment vs. control) at baseline and endline.
2. Channel Cost-Effectiveness: What is the optimal delivery mix between Community FM Radio and personalized 2G GSM robocalls in driving advisory comprehension and adoption?
• *Why Critical:* Calibrates our scaling model to maximize impact per dollar.
• *Method:* Telecom gateway call logs (completion, duration) paired with listener recall surveys and unit-cost accounting.
3. Yield & Welfare Impact: Does advisory compliance translate into measurable 15% to 25% yield gains and net household income increases?
• *Why Critical:* Proves the social return on investment required for public and commercial scaling.
• *Method:* Post-harvest crop-cut yield measurements and farm-gate profit-and-loss surveys. *(1,218 characters)*

Prompt: Impact Evaluation Technical Specifications [up to 1,100 characters]

• **Unit of Assignment:** Rural village clusters (30 clusters; 15 treatment, 15 control) to eliminate localized information spillover.
• **Groups:** Treatment (1,500 smallholders across 15 clusters) receives localized vernacular radio bulletins and direct 2G GSM robocall alerts. Comparison (1,500 smallholders across 15 matched clusters) relies on status-quo extension.
• **Power & MDE:** With N=3,000 smallholders (100/cluster), power=0.80, alpha=0.05, and intra-cluster correlation (ICC)=0.05, the design is powered to detect a Minimum Detectable Effect (MDE) of 0.16 standard deviations—equivalent to detecting an 8 to 10 percentage point gain in advisory adoption or an ~11% yield increase.
• **Data Sources:** Enumerator-administered digital household surveys (baseline/endline), post-harvest crop-cut measurements, and telecom call-completion logs.
• **Timeline:** Months 1–3: Baseline survey; Months 4–15: Two seasonal rollout cycles; Months 16–18: Endline survey and econometric DiD analysis. *(981 characters)*

10. Implementing Organization, Partners & Complete Research Team

Prompt: Organization and key partners [up to 975 characters]

Naysa-84 Labs (Nigeria) leads implementation in consortium with university faculty:
• Dr. Young I. Irivboje (PI, 50% LOE): Systems architect, biometrics & AI engineering.
• Dr. Alaba M. Dare (Co-I, 25% LOE): Ag economics & DiD evaluation lead.
• Dr. Taiwo Ogunseitan (Co-I, 20% LOE): Extension & farmer mobilization lead.
Co-investigators span crop science (Dr. Oyekale), animal breeding (Dr. Sanda), animal science (Dr. O. Irivboje), veterinary health (Dr. Okusanya), and entomology (O. Odumuyiwa, UGA).

Partnerships & Co-Funding:
• *Academic Partners (In Place):* FUNAAB, Ilaro, and OOU contribute faculty time, historical datasets, and field trial plots.
• *Implementation Partners (Pipeline):* State ADPs and community FM stations for field rollout. Africa's Talking telecom gateway is integrated.
• *Co-Funding:* ~$20,000 in-kind software IP and compute infrastructure contributed by Naysa-84 Labs. *(895 characters)*

Complete Research Team Roster (Level of Effort & Roles)

Researcher Name & Degree	Institutional Affiliation	Domain Expertise / Project Role	LOE
Dr. Young Imienwain Irivboje **Lead Investigator**	Naysa-84 Labs, Ogun State / Dept of Animal Breeding & Genetics, FUNAAB *irivboje.youngi@pg.funaab.edu.ng*	**Principal Investigator & Systems Architect:** Overall project direction, biometrics engine (ANOVA, BLUP, GxE, heritability), multi-agent AI copilot & speech models.	**50% FTE**
Dr. Adeyinka Julius Sanda **Senior Co-Investigator**	Dept of Animal Breeding and Genetics, FUNAAB, Ogun State *sandaaj@funaab.edu.ng*	**Quantitative Genetics Specialist:** Livestock genetic biometrics, selection indices, variance components, and tropical climate-resilience breeding strategies.	**15% FTE**
Dr. Alaba Modupe Dare **Senior Co-Investigator**	Dept of Agricultural Economics & Farm Management, OOU, Ago-Iwoye *dare.alaba@oouagoiwoye.edu.ng*	**Agricultural Economics & Impact Evaluation Lead:** Directs quasi-experimental DiD evaluation, baseline/endline household surveys, econometrics, and farm budget analysis.	**25% FTE**
Dr. Taiwo Ogunseitan **Co-Investigator**	Dept of Agricultural Extension Management, Fed. Univ. of Tech., Ilaro *taiwo.ogunseitan@federalpolyilaro.edu.ng*	**Agricultural Extension & Dissemination Lead:** Farmer cluster mobilization, cooperative onboarding, extension feedback loops, and radio advisory pedagogical clarity.	**20% FTE**
Dr. Oyekale Josiah I. **Co-Investigator**	Dept of Crop Science, Faculty of Agriculture, Fed. Univ. of Tech., Ilaro *Josiah.oyekale@federalpolyilaro.edu.ng*	**Crop Science & Agronomy Lead:** Crop trial biometrics, stoichiometric soil fertility (lime & NPK balancing), crop phenology, and planting window validation.	**15% FTE**
Dr. Oluwabusayo A. Irivboje **Co-Investigator**	Dept of Animal Science, Faculty of Agriculture, Fed. Univ. of Tech., Ilaro *olubusayo.irivboje@federalpolyilaro.edu.ng*	**Animal Science & Nutrition Specialist:** Smallholder livestock production analytics, feed formulation, and small ruminant/poultry trial evaluations.	**15% FTE**
Dr. Phebe O. Okusanya (DVM) **Co-Investigator**	Dept of Animal Science, Faculty of Agriculture, Fed. Univ. of Tech., Ilaro *phebe.okusanya@federalpolyilaro.edu.ng*	**Veterinary Medicine & Bioclimatic Specialist:** Temperature-Humidity Index (THI) heat stress models, veterinary disease alert thresholds, and herd health scripts.	**15% FTE**
Oludare E. Odumuyiwa (MS) **Co-Investigator**	Tree Nut & Fruit Entomology Lab, University of Georgia (UGA), USA *Oludare.Odumuyiwa@uga.edu*	**Entomology & Pest Surveillance Specialist:** Pest outbreak early warning models (Fall Armyworm, stem borers) and economic threshold alert calibration.	**10% FTE**

11. Academic & Policy Citations

Prompt: Citations referenced in application responses

1. Cole, S., & Fernando, A. N. (2021). 'Mobile'izing agricultural advice: Technology adoption, diffusion and outcomes. *The Economic Journal*, 131(634), 986–1017. https://doi.org/10.1093/ej/ueaa084
2. Fabregas, R., Kremer, M., & Schilbach, F. (2019). Realizing the potential of digital development: The case of agricultural extension. *Science*, 366(6471), eaay3038. https://doi.org/10.1126/science.aay3038
3. Van Campenhout, B., Spielman, D. J., & Lecoutere, E. (2021). Information and communication technologies to provide agricultural advice to smallholders: Experimental evidence from Uganda. *American Journal of Agricultural Economics*, 103(1), 317–337. https://doi.org/10.1111/ajae.12117
4. Duflo, E., Kremer, M., & Robinson, J. (2011). Nudging farmers to use fertilizer: Theory and experimental evidence from Kenya. *American Economic Review*, 101(6), 2350–2390. https://doi.org/10.1257/aer.101.6.2350
5. Federal Ministry of Agriculture and Food Security (FMAFS / FMARD). (2022). *National Agricultural Technology and Innovation Policy (NATIP) 2022–2027*. Federal Government of Nigeria, Abuja.
6. National Bureau of Statistics (NBS). (2022). *Nigeria Multidimensional Poverty Index & Smallholder Agricultural Survey*. Abuja, Nigeria.
7. World Bank. (2022). *Bridging the Extension Gap: Digital Technologies for Smallholder Productivity in Sub-Saharan Africa*. World Bank Group, Washington, D.C.

PROPOSAL STATUS: Application Complete & Archived	DIV FUND STAGE: Stage 1 (Piloting - $150k)	COLLABORATION: Open for Academic Consortium